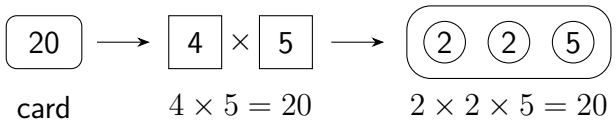
A number card can split into two whole numbers greater than 1 that multiply to its value. These numbers are its factors. Keep splitting until every card is prime: a number greater than 1 that cannot split. The finished cards make a prime bag. Multiply its tokens to find its value.



Token order does not matter. Tokens with the same number are the same.

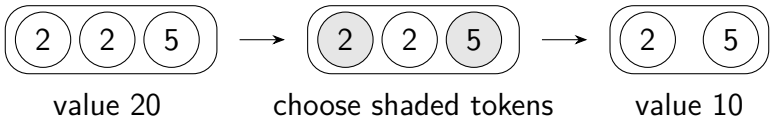
Problem 1: Find different splitting routes for 36, 48 and 60. Can a number have two different finished prime bags?

36

48

60

Choose some tokens from a bag, using each token at most once. The chosen tokens form a bag. You may choose all of them or none. An empty bag has value 1. A divisor of a whole number divides it with no remainder.



value 20 choose shaded tokens value 10

$2 \times 5 = 10$ $20 \div 10 = 2$

Problem 2: For each bag, find every different value you can make by choosing tokens. Explain why your collection is exactly the set of divisors of the full bag's value.

2 2 2 3 value 24

2 3 5 value 30

Problem 3: How many different values can you make by choosing tokens from each bag? Find a way to be certain of each total without listing every value.

(2)(2)(3)(3) value 36

(2)(2)(3)(5) value 60

(2)(2)(2)(3)(3) value 72

Problem 4: A bag has seven 2 tokens and four 3 tokens. How many different divisors does its value have? Explain a counting rule that works for any prime bag.

Problem 5: Draw prime bags using any of 2, 3 and 5 whose values have exactly 6, 8, 9, 10 and 12 divisors. Find all possible token-count patterns for each target. A bag with two 2s and one 3 has the same pattern as a bag with two 5s and one 2.

6 divisors

8 divisors

9 divisors

10 divisors

12 divisors

Problem 6: Find the least positive number with each divisor count below. You may use any primes. Explain how you can rule out every smaller number.

4 divisors

6 divisors

8 divisors

9 divisors

10 divisors

12 divisors